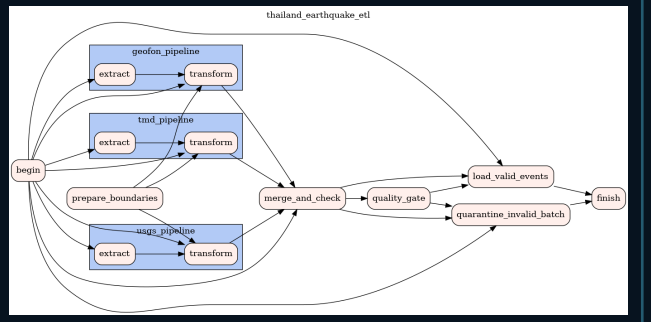
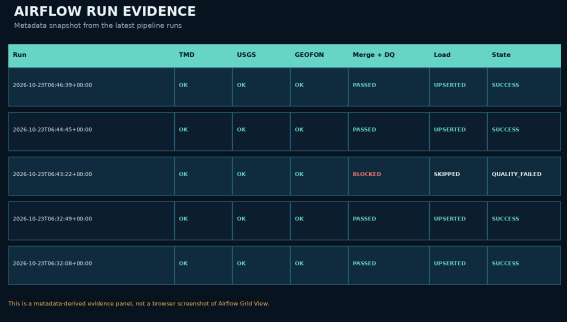
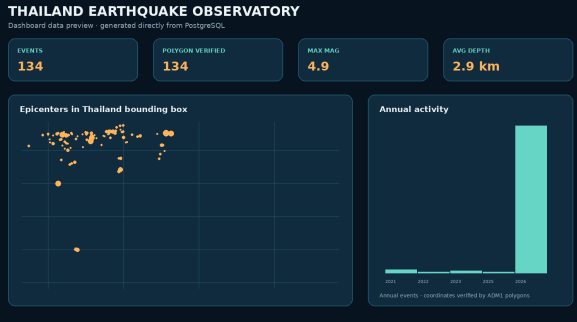
Thailand Earthquake Pipeline · Project overview

Achievement: Level 3 + bonus. Events from 2021-2026 are retained only when their coordinates fall inside a real Thailand ADM1 polygon; the matching polygon supplies the province. Three sources are normalized, deduplicated, quality-checked, and idempotently upserted. The dashboard reads PostgreSQL directly.







Problems found, diagnosis, and resolution

Problem / symptom	How the cause was identified	Resolution
TMD has no machine API and the page mixes Thai local time with UTC.	Inspected the real HTML: event rows have EQ IDs and a second timestamp explicitly marked UTC.	Requests + BeautifulSoup only; parse UTC, honor robots.txt, limit to 5 pages with a 1-second delay, and detect schema drift when zero rows are found.
A place name or rectangular bounding box includes events outside Thailand.	Plotted edge coordinates and found points in Myanmar, Laos, Malaysia, and the Gulf despite matching the rough box.	Use point-in-polygon against geoBoundaries ADM1 geometry; keep only a matching polygon and use its name as the province.
A source outage could fail the entire scheduled DAG.	Failure propagation from parallel extraction tasks would block the downstream merge.	Catch source-level network errors into status records; merge always runs and labels the batch degraded. Quality still controls loading.
Repeated runs could inflate the database.	A plain INSERT treats the same catalog event as new on every schedule.	Stable event fingerprint + PostgreSQL primary key + ON CONFLICT DO UPDATE; source arrays and IDs are merged.

Planning, prioritization, and scope decisions

20%

1 · Foundation

Confirm source contracts, robots/PDPA, 2021-2026 scope, and province polygons.

40%

2 · Reliable ETL

Build parallel TaskGroups, polygon filtering, validation, dedup, and UPSERT tests.

25%

3 · User experience

Create point map, annual counts, magnitude statistics, filters, and status view.

15%

4 · Evidence

Run twice, verify 3-source counts and 100+ exact-scope rows, then finalize report.

Deliberately excluded: Selenium (explicitly prohibited), personal or felt-report respondent data (PDPA risk), predictive claims, and events that merely contain the word Thailand while their coordinates fall outside the country polygons. Priority was given to reproducibility, geographic correctness, failure tolerance, data quality, and idempotency.

Submission checklist: 100+ database rows · run DAG twice · capture live Graph/Grid/dashboard images · confirm second count is unchanged · export this PDF (maximum 3 pages).